Designed to Requirements: When Compliance Stands in for Knowing a Design Is Good

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Abstract

Engineers solve problems by relying on experience, standards, and colleagues. Still, no engineer or engineering team possesses all knowledge and therefore designs, especially novel designs, may be compliant yet flawed. The engineering knowledge gap can be narrowed, though it is sometimes not realised until failure. Engineering is a collaborative activity in which all parties, including individual engineers and design teams, influence design judgement as it progresses. How does the engineering team decide when it has done enough?

In 1988, a Boeing 737 operating as Aloha flight 243 suffered a significant fuselage failure in which part of the roof, from about the main cabin entrance door to just forward of the wings, tore away. The aeroplane had operated short island hops for nearly 20 years. Boeing's testing showed that its design could operate for 150,000 pressurisation cycles and tolerate the expected damage from a failed turbine blade. However, Boeing did not explicitly consider the aircraft's full operating environment, nor did it consider how expected fuselage cracks might join, then tear. One way design engineers can widen the failure modes they consider is by cultivating safety imagination, sharing good and bad design stories, and encouraging and supporting dissent within the ranks.

Engineers should appreciate that hazard identification and risk management need to evolve, and that repeating past methodologies that may have resulted in a successful design could, in the future, lead to design inadequacy. The paper highlights the difficulties engineers face when developing a design and assessing its sufficiency.

Keywords: System safety, Engineering judgement, Design adequacy, Safety standards, Safety assurance

1 Introduction: Searching for Design Inadequacy

When developing a new design, engineers request, further develop and follow design requirements, which are the design's expected outcomes and specified constraints. However, while developing the design, requirements are only one of the design's influences. Practising engineers bring their past experiences and the design principles derived from them to each new design (Ahmed and Christensen 2009). Engineering design is a collaborative activity where engineers from different disciplines and other specialists develop engineered solutions (Trevelyan 2014). No engineer knows everything, nor does a single engineer have infinite competence, as design engineering is a collaborative effort utilising multiple engineering experts across many complementary engineering disciplines.

When engineering novel designs, engineers still draw on their existing experience; however, that experience may be less directly applicable to the new design. The need to develop a novel design may place greater demands on an engineer's judgement, as it will require considering designs beyond those they have previously encountered. Formal design requirements and well-known engineering principles can establish a baseline, but neither can ensure that a design is adequate in a given context.

In this paper, design inadequacy means a safety-related problem that designers did not consider or that they considered but misunderstood. This includes problems omitted from or misrepresented in requirements, as well as problems arising when requirements are interpreted or applied during design. The focus is on design engineers and the contributors to their design judgements, including those undertaking assessment, testing and assurance review. The paper examines how standards, experience, assessment and testing support judgements about whether the search for safety-related design inadequacies has been adequate. It distinguishes finishing the required activities from having good reasons to judge that the search has been thorough enough, and it does not assess whether remaining risks should be accepted or whether a design should be released. Finally, legal responsibilities and the overall adequacy of the safety case also fall outside its scope.

2 Standards, Assurance and Adequacy Judgements

Consistent with the strong focus on codified forms of knowledge in engineering, an accepted method of system safety assurance is to assess compliance with standards. In this approach, an engineered solution is assessed against domain-specific standards, providing confidence that the required safety objectives are met (Nair et al. 2014). For example, in the process industry, safety instrumented systems under IEC 61511; airborne software under DO-178C; railway systems/signalling under the relevant CENELEC standards (EN 50126, EN 50128 and EN 50129); and automotive functional safety, to ISO 26262 (Nair et al. 2014, Smith and Simpson 2016). These standards provide a structured outline for sector-based assurance. Standards may prescribe activities or design features, specify goals, or combine these approaches (Kelly, McDermid and Weaver 2005, Hawkins et al. 2013, Nyvik, Flage and Guikema 2021). Additionally, design assurance can include quantitative measures, such as reliability calculations, while other aspects may be qualitative. The focus on the engineering team and the standards is to provide adequate confidence of safety rather than explicit proof (Nair et al. 2014). While engineers do

their best to provide adequate evidence of compliance with standards, the standards themselves are typically vague about what constitutes adequate evidence (Nair et al. 2014). In many cases, standards are deliberately vague, as this stance allows them to serve as general guidance applicable to many different context-specific yet standards-based solutions. As a result, engineers exercise engineering judgement when developing a given design, determining what evidence is sufficient to support their standards-based assurance argument.

3 Prescriptive and Goal-Based Standards

Looking at standards in more detail, we see that engineering standards can fall into several categories, with prescriptive and performance (goal) based standards two common groups. This is important as the category of standard influences what it means to comply.

A prescriptive standard explicitly prescribes how an engineering design is to achieve a specific outcome. From a safety perspective, this simplifies the design justification: if a design meets the specified requirement, it can be accepted as meeting the required safety level within the standard's scope and assumptions (Hawkins et al. 2013, Nyvik, Flage and Guikema 2021). If a novel system needs to be designed and made safe, older or existing standards may struggle to adequately assure the novel design, as the specific aspects of the standards would not have considered the later or more novel assurance needs (Kelly, McDermid and Weaver 2005, Nyvik, Flage and Guikema 2021). Performance- or goal-based standards help by introducing flexibility and requiring that a design meet a particular goal (or perform in a certain way), as detailed in an applicable standard. This flexibility requires designers to judge which means and evidence are sufficient to satisfy the higher-level safety requirements.

Kelly, McDermid and Weaver (2005) describe an evolving safety case that makes compliance arguments explicit and supports agreement between supplier and customer on the evidence required. Such judgements therefore form part of a documented justification that relevant stakeholders can examine and challenge. The question here is how the adequacy of the search is justified within that process.

An example is introduced through Kelly, McDermid and Weaver (2005) who discuss DS 00-56, Issue 3:

Defence Standard 00-56 Issue 3 includes the following objectives, summarised here:

1) Objective (i): all credible hazards and accidents are to be identified.
2) Objective (j): identified risks are subsequently to be reduced to the specified risk condition.

Clause 10.5.1 requires hazard identification and hazard analysis, while Clause 10.5.2 requires the contractor to demonstrate the adequacy of those processes and the suitability of the techniques employed. Kelly, McDermid and Weaver (2005) explain that the standard deliberately leaves open what is sufficient to meet these goals in a particular context.

This standard's framework drives an engineer to perform hazard identification, then hazard analysis and risk reduction. The design engineering team should have judged the hazard-identification and analysis process adequate, and that deemed adequacy must be defended. The design should also meet the standard's risk reduction objectives, including ALARP. However, what evidence do engineers have to support their judgement that the hazard identification and analysis process was adequate?

4 Experience and Analogical Reasoning

While standards are useful, successful application requires judgement based on experience. As engineers progress from project to project, they gain experience they can draw on to reason and apply to their current project. Dreyfus and Dreyfus (2009) describe the development of expertise as a progression from following explicit rules towards recognising situations and possible responses through accumulated experience. Experts can also question their initial understanding and reconsider their response, particularly when decisions are important and time permits (Dreyfus and Dreyfus 1986). Further, reflecting on case studies can contribute to contextual understanding and the development of expertise, but does not replace practical experience (Flyvbjerg 2006). This suggests that engineers' experience may influence which features they notice, which similarities they recognise, and how they interpret potential design problems.

Ahmed and Christensen (2009) found that the analogies used by experienced designers were different from those relied upon by more novice designers. The novice designers in their study borrowed from a past job, passing this knowledge directly to the next, while experienced designers leaned on past experience to contemplate and solve problems in their current design. This aspect of design, in which experienced designers draw on analogies from past designs to influence later work, is one method engineers can use to identify problems in new designs (Ahmed and Christensen 2009). Ahmed and Christensen (2009) studied adaptive rather than novel designs. Their work can support reasoning from an analogical perspective, but they do not directly show how engineers reason about novel systems. Past knowledge may not be directly transferable to later designs; applying the analogy involves reasoning about the identified similarities between the two designs, how each system may behave, and how appropriate the developed analogy is for the later design.

Driscoll (2010) offers an illustrative estimate of designers' hands-on system experience (20 years into a 40-year career), suggesting they may have only around 5,000 hours of real hands-on system experience. Therefore, while an engineer may have had a long career, direct exposure is limited. When comparing that figure to the required failure-probability limits in his example, 10^{-7} to 10^{-9} for a one-hour exposure, he argues that "We cannot rely on our experience-based intuition to determine whether a failure can happen within required probability

limits"[1]. Limited personal exposure provides a limited basis for judging rare failures. Maslen and Hayes (2022) argue that field experience ("it's the seeing and the feeling") contributes to pipeline engineers' office-based reasoning by developing an embodied understanding of assets and the people who construct and operate them.

5 Safety Activity, Evidence and False Assurance

The concept of risk provides a common framework for developing a judgement about whether a given design is safe enough. The identification of hazards and the assessment of risk are safety-related tasks in the engineering design process. Tied to this process are several activities intended to improve safety, enhance one's safety knowledge, or allow one to demonstrate safety (Rae and Alexander 2017). Design engineers perform qualitative and quantitative assessments throughout the design lifecycle, with assurance processes performed along the way and to verify the final design. The extensive application of hazard identification and risk assessment processes by itself does not establish that the search was complete. Sometimes, however, these techniques can increase an engineering team's confidence in system safety rather than improve the team's knowledge of the risk. Rae and Alexander (2017) call this probative blindness, which is a property of a particular activity in particular circumstances, and is not a claim that any of these hazard and risk management techniques are inherently bad.

Supporting these risk activities are physical testing and simulation activities, with the risk activities informing these activity types. Tests can provide evidence that a component works as expected under specific conditions, whereas simulation provides a means to model expected system behaviour when physical testing would be costly or dangerous. Testing and simulation results depend on simulation and test assumptions and on how adequately relevant operating conditions are represented (Oberkampf and Trucano 2002). There is a trap that engineers can fall into: a successful test or simulation should not provide certainty that there are no underlying system abnormalities, as testing or simulation provides evidence of behaviour under test conditions (Sargent 2008).

6 Widening the Search: Safety Imagination, Stories and Dissent

Established safety techniques based on risk or compliance can restrict inquiry when their assumptions and categories fail to properly appreciate the intended task. An organisation becomes competent at building systems, develops techniques for identifying and managing typical hazards and risks, and continues to produce reasonably successful designs. When novelty enters the mix, these techniques may no longer be suitable, even though the designs continue to function successfully. To combat fixed repetitive processes, Pidgeon and O'Leary (2000) suggest safety imagination as a useful approach. Safety imagination encourages inquiry beyond the assumptions and categories used in an existing hazard assessment. Hazard and risk assessment techniques that embrace a "safety imagination" method drive the engineer to consider beyond the expected. To help promote the method, Pidgeon and O'Leary (2000) offer a checklist for safety imagination guidelines. Squair (2024) describes reverse TRIZ as a supporting technique for hazard identification. It encourages participants to consider how they could make a system fail, helping them explore hazards or causal mechanisms that another analysis may have missed. Such a technique can help design engineers who participate in seemingly repetitive processes gain an alternative lens that could spark a necessary shift in perspective. Safety imagination is a method that widens inquiry. It does not guarantee that the technique will discover all possible failures (Pidgeon and O'Leary 2000).

Hayes and Maslen (2015) suggest another method engineers can use to widen what they can imagine. They discuss how stories of disasters, anomalies, and surprises can draw on other people's experience and alter listeners' perception of what is possible (Hayes and Maslen 2015). Engineering experts possess significant knowledge; however, much of it may not be front of mind. Stories can help remind experts of how systems can fail. Stories can widen an engineer's repertoire of analogies and failure patterns. Stories, however, are an informal way of sharing knowledge.

Safety imagination is a simple concept; however, it is difficult in practice. Ramsden (1984) discussed safety imagination in aircraft maintenance and argued for greater use of confidential incident reporting by maintainers. Existing schemes were available to them, but were used predominantly by pilots and air traffic controllers. Westrum (1993) develops a related organisational argument and argues that imagination is not only useful; it is required, discussing requisite information concerning possible modes of failure as well as possible ways to test them (Westrum 1993). When one considers imagination, it appears, at first glance, to be an individual skill. However, Westrum (1993) argues that whether such imagination is expressed and used depends significantly on organisational culture, and the organisation's ideas need to be encouraged, whether they come from the front line or the back office. The organisation must encourage all of its actors to voice their ideas and concerns; these concerns must be heard; there must be feedback to encourage more of this thinking: "Sound organizations nurture the expression of doubt" (Westrum 1993, p. 407). Squair's (2024) reverse TRIZ example has a parallel in Westrum's (1993) discussion of the "dirty-minded" person who is expert in what can go wrong. Such a person imagines nefarious ways to fail a system or communicates how a system could unexpectedly function. It is parallel to the argument that Leveson (2011) makes when considering problems with safety cases; she suggests they can encourage confirmation bias, and that considering the opposite would be beneficial. Driscoll (2010) recommends including sceptical participants and

[1] The presentation does not explain how this estimate was derived; it is used here as an illustration rather than an empirical estimate of designers' experience.

members of related disciplines in failure modes and effects analysis teams to challenge designers' optimism.

7 Aloha 243: When a Known Failure Mode is Misunderstood

The Aloha case provides a parallel to the above, in which the engineers developed a compliant design through successful testing. Aloha Airlines Flight 243, a Boeing 737 on a milk run between the Hawaiian Islands, suffered a catastrophic fuselage failure due to cracking on April 28, 1988 (NTSB 1989, Downer 2011). A large section of the upper fuselage separated and the cabin decompressed. One person died, but the aircraft landed. The accident occurred due to unconstrained cracking of the fuselage along a joint. Disbonding of a lap joint transfers loads to the rivets and can promote fatigue cracking at the rivet holes. Multiple-site damage (MSD) comprises multiple fatigue cracks along a rivet line. The NTSB concluded that numerous cracks linked rapidly during the accident, causing catastrophic failure of a large section of the fuselage (NTSB 1989).

During its initial development, the Boeing 737 underwent extensive testing; however, the laboratory testing represented a subset of what could be experienced in service. The original fatigue test subjected a representative crown-to-keel fuselage half-section to 150,000 pressurisation cycles; no fatigue cracks or disbonding developed. These results supported the aircraft's fatigue-life expectations, but the fatigue evaluation did not consider possible disbonding or the effects of corrosion on fuselage lap-joint strength (NTSB 1989).

To support its own fail-safe design requirements, Boeing conducted a test to determine whether the fuselage could withstand a 40-inch crack, the maximum expected external damage from an uncontained engine failure. This represented a localised rupture rather than distributed fatigue damage; Boeing's test was intended to demonstrate that the fuselage could fail without catastrophically rupturing. The fuselage design incorporated tear straps to redirect cracks, and the guillotine test demonstrated that fuselage shear was controlled. Once sheared, the fuselage skin tore; however, it was constrained by the tear straps with controlled depressurisation and the sheared skin flapping safely. The intended crack redirection depended on the integrity of the structure ahead of the crack; sufficient fatigue cracking, tear-strap disbonding, or both could defeat controlled decompression (NTSB 1989).

The guillotine test was not the only test performed. The design engineers also considered pressurisation cycles, which was important for an aircraft such as the Boeing 737 used on flight 243 (Downer 2011). Flight 243 was an early Boeing 737 entering service in 1969 and was used for short hops, thus being exposed to multiple pressurisation cycles (NTSB 1989). There were aspects of the aircraft that would remain untested; for example, it was used in a warm environment and exposed to saltwater (Downer 2011). The Boeing 737 pressurisation test relied in part on past Boeing 727 tests; therefore, Boeing deemed a half-section pressurisation test adequate (NTSB 1989). Additionally, these tests were conducted in an environment far more sterile than the one in which Aloha 243 operated (Downer 2011). The NTSB identified Aloha Airlines' failure to detect significant disbonding and fatigue damage through its maintenance programme as the probable cause of the accident (NTSB 1989).

At the time of the Boeing 737's initial certification, consideration of MSD was not part of the certification requirements (NTSB 1989). The NTSB found that accelerated fatigue testing had not accounted for the effects of the in-service environment on bonded lap joints and tear straps (NTSB 1989). The Aloha case illustrates why successful testing must be interpreted in relation to the conditions and damage mechanisms represented in the tests. Meeting the certification requirements provided evidence supporting the design, but did not establish that the effects of deterioration in service had been adequately understood. When the assessor has the results of the half-section fuselage test, how do they know it can truly represent the operating life of an equivalent aircraft?

8 What Makes the Search Adequate?

Revisiting DS 00-56, this standard requires that engineering designers identify all credible hazards. The identification must be adequate, and there is wide variation in technologies and therefore in the determination of sufficiency (Kelly, McDermid and Weaver 2005). Objective (j) of DS 00-56 Issue 3, summarised earlier, requires reduction of the risks associated with identified hazards, which parallels the "So Far As Is Reasonably Practicable" (SFAIRP) requirement of risk management. DS 00-56 uses the term "As Low As is Reasonably Practicable" (ALARP), which the UK Ministry of Defence describes as broadly equivalent (ASEMS n.d.). Evidence that the risks associated with identified hazards have been adequately reduced does not, by itself, establish that the search for hazards was sufficiently thorough. Nair et al. (2014) support that in large systems, explicit evidence of safety does not amount to definitive proof; such systems rely on sufficiency and the evidence that supports confidence. Additionally, Nair *et al.* (2014) identify quantitative, qualitative, checklist-based and logic-based approaches for assessing the adequacy of evidence. Ahmed and Christensen (2009) show how experience supports problem identification and reasoning among the designers they studied; however, their study does not establish a criterion for determining when a search for safety-related design inadequacies is sufficient. Hayes and Maslen (2015) help widen designers' considerations by reflecting on how stories can open up a designer's thoughts. Experts are reminded through stories that their knowledge is finite, and that their perception shift may help them build on their "knowledge bank" (Hayes and Maslen 2015), passing and transforming ideas between people. Pidgeon and O'Leary (2000) widen the frame further as they argue that some critical errors are latent or misunderstood. They suggest that the engineer use "safety imagination" to look beyond the problem's constrained framing (and provide a table of ideas to assist), though it has its limits. Westrum (1993) emphasises organisational conditions that encourage people to raise concerns and ensure information reaches those able to act on it, regardless of the contributor's position or status. These approaches can strengthen the grounds for judging a search adequate, but applying them

does not automatically establish its adequacy in a particular design context.

9 Conclusion

Engineers draw confidence from their experience and from evidence that a design meets its expected outcomes and specified constraints. However, evolving capability may weaken the applicability of past experience. In this paper, we have seen that assurance processes are laden with judgement calls and that an experienced engineer is a very useful contributor to a design, but experience also has its caveats. Experienced design engineers can benefit from a broad imagination and are further aided by group processes that encourage dissent. Safety imagination, stories and dissent help engineers investigate potential design problems, but engineers must still judge whether the search has been sufficiently thorough. Successful test results inform that judgement, with their relevance depending on how well the test conditions and assumptions represent the system's intended use. We need to know what engineers rely on when they decide their adequacy search is closed, and how that judgement is made across the design, checking, and analysis parts of a system's lifecycle. Widening the search may expose issues that would otherwise remain outside the frame, but it does not necessarily prevent engineers from misunderstanding issues already within it. Being challenged through the methods discussed may widen inquiry and challenge existing assumptions. Engineering design is not a clearly bounded task, so how does the engineering team decide when it has done enough?

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